
Keeping the Model Accurate as Cheaters Adapt

To maintain accuracy against evolving spoofing techniques, the model should be periodically retrained using newly collected examples of screen photographs captured under different conditions, including varying brightness levels, display technologies (LCD/OLED), viewing angles, reflections, and camera devices. Hard examples where the model makes incorrect predictions should be added back into the training dataset (active learning). Regular retraining ensures the model adapts to new attack patterns over time.

Making the Model Tiny and Fast for Mobile Deployment

Although MobileNetV3-Large is already optimized for mobile devices, the model can be made even smaller and faster by using MobileNetV3-Small, model quantization (INT8), and pruning. Converting the model to TensorFlow Lite or ONNX would further reduce inference time and memory usage, enabling real-time execution on smartphones with minimal battery consumption.

Choosing the Fraud Detection Threshold

The model outputs a probability (fraud score) between 0 and 1. Instead of always using a threshold of 0.5, the threshold should be selected using a validation dataset by analyzing the trade-off between false positives and false negatives. Depending on the application, the threshold can be increased (e.g., 0.7) to reduce false fraud alerts or decreased to detect more suspicious images. The optimal threshold would be chosen based on validation metrics such as precision, recall, F1-score, or ROC analysis.